

Course outline (Syllabus of the course)

Course Code: CS1109 Name of the Programme: BCA (Hons) Name of the Course: Computer Organization Semester: II		
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	45 + 0 + 0	45
Course Objectives:		
a) Understand the fundamental components and operations of computer systems.		
b) Analyze instruction sets and addressing modes.		
c) Explore memory hierarchy, storage systems, and input/output mechanisms, including buses and I/O interfacing.		
d) Comprehend basic concepts of instruction pipelining and overall system performance enhancement techniques.		

Module - 1	9 Hrs
FUNDAMENTAL BLOCKS OF COMPUTER: Computer Types, Functional Units, CPU, I/O subsystems, memory, control unit COMPUTER ARITHMETIC: Signed number representation, Fixed and Floating point representations.	
Module - 2	9 Hrs
INSTRUCTION SET ARCHITECTURE: Memory Location and addresses, Memory Operations, Instruction format, addressing modes, assembly languages, Complex Instruction Set Computer (CISC), Reduced Instruction Set Computer (RISC), CISC vs RISC.	
Module - 3	9 Hrs

MEMORY HIERARCHY: Concept of hierarchical memory organization, Cache Memory- Mapping Functions and Replacement Algorithms, Examples of Mapping Techniques, Cache Coherence.	
INPUT OUTPUT AND SECONDARY STORAGE: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA, Secondary Storage.	
Module - 4	9 Hrs
ALU, DATA-PATH AND CONTROL UNIT: Control Unit and design, Hardwired v/s Micro-programmed Control Unit, Horizontal micro-programmed Vs Vertical micro-programmed control unit, Synchronous Data Transfer, Asynchronous Data Transfer	
Module - 5	9 Hrs
INSTRUCTION PIPELINING: Pipelining, Types of pipelining, Pipelining - Execution, Stages and Throughput, Pipelining - Dependencies and Data Hazard, Branch Delays, Pipelining - Types and Stalling Different Instruction Cycles, Performance of Computer.	

Course Outcomes:	
a)	Describe the functional organization of a computer system and perform arithmetic operations using signed, fixed-point, and floating-point number representations.
b)	Analyze instruction set architectures, addressing modes, and instruction formats, and distinguish between CISC and RISC architectures.
c)	Evaluate memory hierarchy concepts, cache mapping techniques, replacement algorithms, cache coherence, and various I/O and secondary storage mechanisms.
d)	Explain the design and functioning of ALU, datapath, and control units, and analyze pipelined execution, hazards, and performance metrics of computer systems.

Textbooks	
1	Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 6th edition, McGraw Hill, New Delhi, India. ISBN: 9789355323729

Reference Books	
1	M. Mano, Computer System Architecture, 3rd Ed., Pearson, 1992. ISBN: 9789332585607

2	William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey. ISBN: 9788131732458
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Reference Links	
1	https://www.geeksforgeeks.org/computer-organization-and-architecture-tutorials/
2	https://www.javatpoint.com/computer-organization-and-architecture-tutorial
3	https://www.studytonight.com/computer-architecture/

Lesson plan

Name of the Programme:	BCA	
Title of the Course:	Computer Organization	
Course code	CS1109	
Semester:	2	
Names of the Course Instructor(s)	Prof. Kaipangal Maneesh Rajan and Prof. Sharath BR	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	45 + 0 + 0	45

Session	Module No.	Topic	Pedagogy / Activities	Pre-reading / Reference
1	M O D U L E I	Introduction to Computer Organization & Computer Types	Interactive lecture, real-life examples	Hamacher Ch.1, GFG Intro
2		Functional Units of a Computer	Chalk & talk, block diagram discussion	Hamacher Ch.1
3		CPU Components and Functions	Diagram-based explanation	Mano Ch.1
4		I/O Subsystems Overview	Case study: keyboard, printer	GFG I/O basics
5		Memory Unit & Control Unit	PPT + concept mapping	Hamacher Ch.1
6		Signed Number Representation	Numerical problem solving	Mano Ch.2
7		Fixed Point Representation	Worked examples	Hamacher Ch.2
8		Floating Point Representation (IEEE)	Board examples + practice	Stallings Ch.3
9		Numerical Problems & Recap	Tutorial + quiz	Previous references

10	M O D U L E II	Memory Locations & Addresses	Lecture with memory model	Hamacher Ch.3
11		Memory Operations	Illustration + examples	Mano Ch.4
12		Instruction Format	Instruction breakdown	Stallings Ch.11
13		Addressing Modes – Part I	Example-based teaching	Hamacher Ch.3
14		Addressing Modes – Part II	Problem solving	GFG Addressing Modes
15		Assembly Language Basics	Demo assembly snippets	Mano Ch.4
16		CISC Architecture	Comparative lecture	Stallings Ch.14
17		RISC Architecture	Discussion + examples	Hamacher Ch.15
18		CISC vs RISC	Debate & summary	All references
19	M O D U L E III	Memory Hierarchy Concept	Diagram-based lecture	Hamacher Ch.5
20		Cache Memory Basics	PPT + cache analogy	Stallings Ch.4
21		Cache Mapping Techniques	Numerical examples	Mano Ch.6
22		Cache Replacement Algorithms	Comparison activity	GFG Cache
23		Cache Coherence	Conceptual explanation	Stallings Ch.6
24		I/O Interface	Lecture + device model	Hamacher Ch.7
25		Programmed & Memory Mapped I/O	Flowchart explanation	Mano Ch.7
26		Interrupt Driven I/O & DMA	Case study	Stallings Ch.7

27		Secondary Storage	Video + discussion	StudyTonight
28	M O D U L E IV	ALU – Functions & Design	Diagram + examples	Mano Ch.10
29		Datapath Components	Block diagram walkthrough	Hamacher Ch.6
30		Control Unit Overview	PPT explanation	Stallings Ch.13
31		Hardwired Control Unit	Comparative analysis	Mano Ch.11
32		Micro-programmed Control Unit	Microinstruction demo	Hamacher Ch.8
33		Horizontal vs Vertical Microprogramming	Table comparison	Stallings
34		Synchronous Data Transfer	Timing diagram	Mano
35		Asynchronous Data Transfer	Handshaking example	GFG
36		Recap & Problem Solving	Tutorial session	All references
37	M O D U L E V	Introduction to Pipelining	Real-life analogy	Hamacher Ch.14
38		Types of Pipelining	Diagram-based teaching	Stallings
39		Pipeline Stages & Throughput	Numerical problems	Mano
40		Pipeline Execution	Worked examples	Hamacher
41		Data Hazards	Case analysis	GFG
42		Control Hazards & Branch Delays	Problem discussion	Stallings
43		Pipeline Stalling	Diagram + explanation	Mano
44		Different Instruction Cycles	Comparative lecture	Hamacher
45		Performance Evaluation & Course Wrap-up	Quiz + recap	All references

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component
- CIE 1 (Multiple Choice Question[MCQ] + Multiple Select Question [MSQ])	10 marks out of 20
- CIE 1 (Scenario Based)	10 marks out of 20
- CIE 2	25 marks out of 25
- CIE 3 (Written Test)	15 marks out of 25
- CIE 3 (Presentation+Viva)	10 marks out of 25

Semester End Exams (SEE)	
Mode of Exam	Pen and Paper
Weightage of exam	30%

Course Outcome	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
C01	3	1	2		2	-	-	-	-	-	-	-
C02	3	2	3		2	-	-	-	-	-	-	-
C03	3	2	3	1	2	-	1	-	-	-	-	-
C04	3	2	3	2	2	-	2	-	-	-	-	-
C05	2	1	2		3	-	2	-	-	-	-	-

Rubrics for evaluation of CIE (separate for each CIE component)

Programme Outcome	1. Complex Problem-Solving and Critical Thinking Graduates should be able to solve different kinds of problems in familiar and unfamiliar contexts, and apply their learning to real-life situations. 2. Creative and Effective Communication Graduates should be able to demonstrate creativity by thinking in diverse ways, dealing with complex problems, innovating, and viewing situations from multiple perspectives. . 3. Analytical and Research Competence Graduates should be able to evaluate the reliability and relevance of evidence, identify logical flaws in arguments, and analyze and synthesize data from various sources to draw valid conclusions and support them with evidence. 4. Teamwork and Leadership Graduates should be able to work effectively and respectfully with diverse teams, facilitating cooperative efforts and acting together towards common goals. 5. Lifelong Learning and Digital Proficiency Graduates should demonstrate the ability to acquire new knowledge and skills for lifelong learning, adapt to changing workplace demands, work independently, and manage time and resources effectively. 6. Multicultural Competence and Ethical Values Graduates should demonstrate knowledge of multiple cultures, interact respectfully with diverse groups, lead diverse teams, and adopt a gender-neutral and empathetic approach. 7. Responsible Autonomy and Environmental Stewardship Graduates should demonstrate the ability to apply knowledge and skills independently, manage projects to completion, exercise responsibility, and ensure workplace safety and security. 8. Community Engagement and Empathy Graduates should demonstrate the capability to participate in community-engaged services and activities to promote societal well-being, and the ability to empathize by understanding the perspectives, experiences, and emotions of others. 9. Global Perspective & Cultural awareness Acquire knowledge about the values and beliefs of diverse cultures, effectively engaging with empathy. 10. Ethical Reasoning & Social Responsibility Demonstrate moral and value-based principles while being socially conscious and accountable. 11. Civic Engagement & Community Service Participate actively in civic life exhibiting a commitment to community service, social justice and the common good. 12. Environmental Sustainability Show responsibility to manage and work towards achieving the Sustainable Development Goals (SDGs).
Total Marks	70

CIE components	MCQ Evaluation (10 Marks) Marking Rule: • Correct Answer: 2 marks • Wrong Answer: 0 marks • No partial marks					
CIE-1 Graded Component 1 (Quiz) Weightage: 10						
CIE components	CO-PO					
CIE-1: Scenario Based Weightage: 10	CO1 PO1, PO2	Rubrics Scenario Based Questions:				
		Criteria	Excellent (7–10 marks)	Good (4–6 marks)	Satisfactory (2–4 marks)	
		Understanding of Scenario (2 marks)	Demonstrates deep understanding; clearly identifies core issues and implications (2 marks)	Shows good understanding; identifies most key issues (1 mark)	Basic understanding; some issues are identified, but lacks depth (0 mark)	
		Application of Concepts (2 marks)	Applies relevant concepts/theories accurately and insightfully(2 marks)	Applies most relevant concepts with minor errors (1 mark)	Attempts to apply concepts, but with some misunderstandings or omissions (0 mark)	
		Problem Solving / Decision Making (2 marks)	Presents well-structured, feasible, and innovative solutions/decisions (2 marks)	Provides workable solutions with reasonable justification (1 mark)	Solutions are simplistic or partially justified (0 mark)	
		Critical Thinking / Justification (2 marks)	Arguments are logical, well-structured, and critically analyzed with evidence/examples (2 marks)	Reasonable logic and structure; some critical analysis and justification (1 mark)	Limited logic and critical thinking; basic justification provided (0 mark)	
		Clarity and Presentation (2 marks)	Clear, concise, well-organized, and free from major errors (2 marks)	Mostly clear and organized; minor errors (1 mark)	Understandable but lacks clarity or organization (0 mark)	
CIE components	CO-PO	Continuous Internal Evaluation-2 (25 Marks)				
CIE-2 Mid Sem Examination (Theory)	CO1, CO2, CO3 PO1, PO2,	Mid Sem Examination (Theory)				

Weightage: 25	PO3, PO4	<table><tr><th colspan="3">Mid Sem Assessment Pattern: 25 Marks</th></tr><tr><th>Sl#</th><th colspan="2">Content</th></tr><tr><td colspan="3">Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)</td></tr><tr><td>1</td><td>5 Questions of 1 mark each</td><td>5</td></tr><tr><td colspan="3">Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)</td></tr><tr><td>2</td><td>4 Questions of 5 marks each with 2 or 3 sub divisions possibly</td><td>20</td></tr></table>	Mid Sem Assessment Pattern: 25 Marks			Sl#	Content		Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)			1	5 Questions of 1 mark each	5	Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)			2	4 Questions of 5 marks each with 2 or 3 sub divisions possibly	20																																
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CIE-3 (Written) Weightage: 15	CO4 - CO5 PO1,PO2,PO4,PO5,PO8	<table><tr><th colspan="5">Question-wise Evaluation Rubric</th></tr><tr><th>Question</th><th>Part</th><th>Criteria</th><th>Description</th><th>Marks</th></tr><tr><td>Q1 / Q2 / Q3</td><td>(a)</td><td>Concept Accuracy</td><td>Correct definition / principle / key idea</td><td>1</td></tr><tr><td></td><td></td><td>Clarity & Terminology</td><td>Clear explanation with proper technical terms</td><td>1</td></tr><tr><td></td><td></td><td colspan="2">Subtotal – Part (a)</td><td>2</td></tr><tr><td></td><td>(b)</td><td>Approach / Method</td><td>Correct procedure / logic / explanation</td><td>1</td></tr><tr><td></td><td></td><td>Working / Steps / Diagram</td><td>Logical steps or neat labeled diagram</td><td>1</td></tr><tr><td></td><td></td><td>Final Answer / Conclusion</td><td>Correct result with justification</td><td>1</td></tr><tr><td></td><td></td><td colspan="2">Subtotal – Part (b)</td><td>3</td></tr><tr><td></td><td></td><td colspan="2">Total per Question</td><td>5</td></tr></table>	Question-wise Evaluation Rubric					Question	Part	Criteria	Description	Marks	Q1 / Q2 / Q3	(a)	Concept Accuracy	Correct definition / principle / key idea	1			Clarity & Terminology	Clear explanation with proper technical terms	1			Subtotal – Part (a)		2		(b)	Approach / Method	Correct procedure / logic / explanation	1			Working / Steps / Diagram	Logical steps or neat labeled diagram	1			Final Answer / Conclusion	Correct result with justification	1			Subtotal – Part (b)		3			Total per Question		5
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CIE-3(Video Presentation) Weightage: 10	CO4 - CO5 PO1,PO2,PO4,PO5,PO8	Rubric Table – Video Presentation (Computer Organization)					
		Max Marks: 10 Group of 3 10 minutes total					
		Criteria	Excellent	Good	Satisfactory	Poor	Max Marks
		Conceptual Understanding	Accurate, in-depth understanding (3 Marks)	Minor errors, mostly clear (2 Marks)	Basic understanding (1 Mark)	Incorrect or unclear (0 Mark)	3
		Content Organization & Time Management	Well structured, within time (2 Marks)	Minor time issues (1.5 Marks)	Weak organization (1 Mark)	Disorganized / over time (0 Mark)	2
		Individual Contribution	Equal participation by all (2 Mark)	Slight imbalance (1.5 Mark)	Unequal participation (1 Mark)	One or more absent (0 Mark)	2
		Presentation & Communication Skills	Clear; confident, technical (2 Mark)	Mostly clear (1.5 Mark)	Average delivery (1 Mark)	Poor clarity (0 Mark)	2
		Visual Aids & Video Quality	1.0 – High-quality visuals (1 Mark)	0.75 – Good visuals (0.75Mark)	0.5 – Basic visuals (0.5 Mark)	0 – Poor/no visuals (0 Mark)	1
Total						10	